Mid Term Odd Semester Examination, Session 2023-24
M. Tech. Part I (Ist Semester) & Ph. D. Course Work
Mining Engineering
Subject: Geo-informatics, Subject Code: MN542

Time: 2 hours **Full Marks: 30**

Note: ***1. Answer all questions.***
2. Make all suitable assumptions wherever necessary.

1. Do as directed:

(a) A digital image having a radiometric resolution of 8 bits, _____ is the maximum value and _____ is the minimum value of the DN that image could represent. **[1]**

(b) If the IFOV for all pixels of a scanner stays constant, it means that _____ resolution will vary from the image centre to the _____ edge. **[1]**

(c) Differentiate between Active and Passive Sensors? **[1]**

(d) If all visible wavelengths are reflected from an object, it will appear _____, while an object absorbing all visible wavelengths will appear _____. **[1]**

(e) What is the most apparent source of electromagnetic energy that you can think of? What "remote sensing device" do you use to detect this energy? **[1]**

2. What type of wavelength and sensor would be the best to monitor the general health of all vegetation cover over an overburden dump for several months, and why? How will you minimise errors in the prediction? **[5]**

3. A company wants to start a surface mine in a mountainous region. Limiting geometric distortions as much as possible, would you choose a satellite-based or aircraft-based scanning system? Explain in terms of imaging geometry. **[5]**

4. **(a)** If you wanted to map the deciduous (e.g., maple, birch) and the coniferous (e.g., pine, fir, spruce) trees in a forest in summer using remote sensing data, what would be the best way to go about this and why? Use the reflectance curves illustrating the spectral response patterns of these two categories to help explain your answer. **[2.5]**

(b) If the spectral range of the 400 channels of the CASI (Compact Airborne Spectrographic Imager) is precisely 0.40 μm to 0.90 μm and each band covers a wavelength of 1.25 nm, will there be any overlap between the bands? **[2.5]**

5. **(a)** How can the LISS-IV sensor operate in both multi-spectral and mono mode? Explain the specifications of the sensor. What is the difference between the IRS-1D and Resourcesat-I LISS III sensors? **[2.5]**

(b) What are the differences between supervised and unsupervised classification? **[2.5]**

P.T.O.

6. **(a)** Define Spectral reflectance. Discuss its importance in the determination of the contamination level of the water. **[2]**

(b) Take a look at the aerial photograph above. Identify race track, river, roads, bridges, residential area and dam in the image and explain the way you are able to do so. **[3]**



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